

SUPPORTING GUIDE · KM-04 + KM-05 INTEGRATED PROJECT

GitHub & VS Code Setup Guide

How to turn your project folder into evidence-grade version control

Occupational Certificate: Artificial Intelligence Software Developer · NQF Level 4



You write your code in VS Code, Git tracks every change you make, GitHub stores your project online as your portfolio evidence, and you submit the GitHub link alongside your zip file on UpSkill.

Why This Matters

GitHub is the industry-standard place for developers to store and share code. Every serious software job application asks for a GitHub profile. Starting this habit now — with small student projects — means that by the time you graduate, you already have a portfolio of real, dated, working projects that any employer can see.

For this project specifically:

Your commit history is part of your assessment evidence. A repository with one commit at 11:58pm the night before the deadline tells a very different story to one with eight or more commits spread across several days, each with a clear message describing what you did.

Step 1 — Create a GitHub Account

1. Go to <https://github.com> in your browser.
2. Click Sign up in the top-right corner.
3. Enter your email address, create a password, and choose a username. Choose a professional username — employers will see it (e.g. a version of your real name works well).
4. Verify your email address using the code or link GitHub sends you.
5. When asked about a plan, choose the Free plan — it's all you need for this project and for almost everything you'll do during this qualification.

Step 2 — Install Git on Your Computer

Git is the version-control tool that tracks your changes. GitHub is the website that hosts your repository online — they are two different things that work together.

1. Go to <https://git-scm.com/downloads> and download the installer for your operating system (Windows, macOS or Linux).
2. Run the installer, accepting the default options — the defaults work fine for this project.
3. Confirm the install worked by opening a terminal (or Command Prompt / PowerShell on Windows) and typing:

```
git --version
```

You should see something like git version 2.43.0. If you see 'command not found', restart your computer and try again — this is usually a PATH issue that a restart resolves.

Step 3 — Set Up VS Code for Git & GitHub

1. If you don't already have it, download VS Code from <https://code.visualstudio.com> and install it.
2. Open VS Code, click the Extensions icon in the left sidebar (four squares icon).
3. Search for 'GitHub Pull Requests and Issues' and click Install. This lets VS Code talk directly to your GitHub account.
4. Search for 'Python' (by Microsoft) and install that too, if you haven't already — you'll need it for this project.

VS Code actually has Git built in already (the Source Control icon, third from the top in the left sidebar) — the extension above just adds GitHub-specific conveniences like signing in and creating repositories directly from VS Code.

Step 4 — Tell Git Who You Are

Git needs to know your name and email so every change you make is correctly labelled. Open a terminal (in VS Code: Terminal menu → New Terminal) and run:

```
git config --global user.name "Your Full Name"
git config --global user.email "your.email@example.com"
```

Use the same email address you used to sign up for GitHub, so your commits are correctly linked to your profile.

Step 5 — Create Your Project Repository

You have two options. Option A is the simplest for most students.

Option A — Create the repository on GitHub first, then clone it

1. On github.com, click the + icon top-right, then New repository.

2. Name it something clear, e.g. km04-05-thabo-corner-store.
3. Choose Public (recommended, so it's easy to share as evidence) or Private if you prefer — either is acceptable evidence as long as you share the link with your facilitator.
4. Tick 'Add a README file', then click Create repository.
5. On the new repository page, click the green Code button, and copy the HTTPS URL (it looks like <https://github.com/yourusername/km04-05-thabo-corner-store.git>).
6. In VS Code, open the Command Palette (Ctrl+Shift+P on Windows/Linux, Cmd+Shift+P on Mac), type 'Git: Clone', paste the URL, and choose a folder on your computer to save it in.
7. VS Code will ask to open the cloned folder — click Open.

Option B — Start from your existing project folder

1. Open your existing project folder in VS Code (File → Open Folder).
2. Open a terminal in that folder and run:

```
git init
git add .
git commit -m "Initial commit: project scaffold"
```

3. On github.com, create a New repository as above, but this time do NOT tick 'Add a README file' (your folder already has content).
4. Copy the commands GitHub shows you under '...or push an existing repository from the command line' and run them in your terminal — they will look similar to:

```
git remote add origin https://github.com/yourusername/km04-05-thabo-corner-store.git
git branch -M main
git push -u origin main
```

Step 6 — Copy Your Project Files In

Whichever option you used above, make sure your cloned/initialised folder ends up containing:

- Your completed starter_code/ folder
- Your data/ folder (or a note in your README explaining where to source it, if your facilitator asks you not to re-upload the raw data — check with them)
- Your PLAN.md and README.md files

Step 7 — The Save-Snapshot-Upload Cycle: Add, Commit, Push

Every time you make progress, repeat this three-step cycle. Think of it as: stage your changes, save a labelled snapshot, then upload it.

Using the VS Code interface (recommended for beginners)

1. Click the Source Control icon in the left sidebar (it shows a number badge when you have unsaved changes).
2. Review the list of changed files. Hover over 'Changes' and click the + icon to stage all of them (or stage files individually).
3. Type a clear, specific message in the message box at the top, e.g. 'Complete data cleaning for products and sales'.
4. Click the checkmark (✓ Commit) button.
5. Click ... (More Actions) → Push, or click the sync icon at the bottom of the window, to upload your commit to GitHub.

Using the terminal (good to know either way)

```
git add .  
git commit -m "Complete data cleaning for products and sales"  
git push
```

Commit often:

Aim for at least 8 separate commits across the eight tasks in the Project Brief — ideally more. Small, frequent commits with honest messages are exactly what assessors want to see, and they also make it much easier for you to undo a mistake later.

Step 8 — Add a .gitignore File

A `.gitignore` file tells Git which files NOT to track — usually temporary files or your virtual environment, which are large and don't need to be shared. In your project's root folder, create a file named exactly `.gitignore` containing:

```
venv/  
__pycache__/  
*.pyc  
.DS_Store
```

Save it, then add, commit and push it like any other change.

Step 9 — Verify Your Repository Online

1. Go to your repository page on github.com.
2. Confirm you can see all your project files listed.
3. Click the Commits link (usually near the top) and confirm you can see your full history of commits with their messages and timestamps.
4. Copy the URL from your browser's address bar — this is the link you will submit alongside your zip file (see Section 8 of the Project Brief).

Troubleshooting Common Issues

"Support for password authentication was removed"

GitHub no longer accepts your account password when pushing from the terminal. When prompted for a password, you need a Personal Access Token instead: on GitHub, go to Settings → Developer settings → Personal access tokens → Generate new token, give it 'repo' permissions, copy it, and paste it in place of your password when Git asks. VS Code's built-in GitHub sign-in avoids this issue entirely — use Option A from Step 5 where possible.

"git: command not found"

Git isn't installed, or your computer hasn't picked up the change yet. Reinstall Git from git-scm.com, then fully restart your computer (not just VS Code) and try again.

"fatal: not a git repository"

You're running a Git command outside your project folder. Make sure your terminal's current folder is the one where you ran git init or git clone.

I committed my venv/ folder by mistake

Add venv/ to your .gitignore as shown in Step 8, then run: `git rm -r --cached venv`, followed by a normal add, commit and push. This removes it from future tracking without deleting it from your computer.

Quick Command Reference

Command	What it does
<code>git status</code>	Show what's changed since your last commit
<code>git add .</code>	Stage all changed files, ready to commit
<code>git commit -m "message"</code>	Save a labelled snapshot of your staged changes
<code>git push</code>	Upload your commits to GitHub
<code>git pull</code>	Download the latest changes from GitHub
<code>git log</code>	View your commit history
<code>git clone <url></code>	Copy a GitHub repository to your computer

You're all set:

Once your repository is live and your commit history reflects the eight project tasks, you have exactly the evidence described in Section 7 of the Project Brief. Copy your repository's URL into your UpSkill submission alongside your zip file.